

Data Mesh Domain Discovery Pipeline (Expert Validation)

We propose an automated pipeline that, when migrating a data lake to datamesh, to infer business domains and generate a human-readable label for each. This questionnaire asks you, as a domain expert, to assess whether the

inferred domains and labels are coherent, well-separated, accurately labeled, and appropriately sized for real-world use.

Out of 10 provided data lakes, please, analyze at least 5. You are provided with two types of entity-relationship diagrams (ERDs): (1) a diagram of the full database schema, showing all tables and their relationships, and (2) a separate diagram for each inferred business domain, showing only the tables assigned to that domain. Please review the full schema diagram first to understand the overall structure of the data lake, then examine each per-domain diagram together with its assigned domain label. **Considering all analyzed selected 5, out of 10, data lakes**, please answer the following questions about the quality of the inferred domains and their labels.

* Indica una domanda obbligatoria

1. Please, Insert your name and surname.

2. Please indicate your current organizational role or profession (e.g., Professor, Post-doctoral Researcher, Software Engineer, Data Architect, Software Developer, Consultant).

3. Please provide your institutional or professional email address.

4. How would you rate your level of expertise in Data Management, Data Architecture, or a related field (e.g., database design, data modeling, data engineering, data governance, business intelligence, enterprise architecture, or software architecture)? *

Contrassegna solo un ovale.

- ☐ Novice
- ☐ Intermediate
- ☐ Advanced
- ☐ Expert

5. How much experience do you have interpreting database schemas or entity-relationship diagrams (ERDs) to infer the underlying business processes they represent? *

Contrassegna solo un ovale.

- ☐ No experience
- ☐ Limited experience
- ☐ Moderate experience
- ☐ Extensive experience

6. **Domain coherence:** A business domain is considered coherent if all its tables contribute to the same business capability. A business capability is a stable, purpose-driven set of activities and information that supports a specific organizational outcome (e.g., managing customer orders, tracking inventory, or processing payments). On average, considering all selected data lake, approximately what proportion of the tables in each inferred domain clearly support the same business capability?

*

Contrassegna solo un ovale.

- ☐ Very poor (Fewer than 25% of tables in each domain clearly support the same business capability)
- ☐ Poor (Roughly 25–49% of tables clearly support the same business capability)
- ☐ Fair (Roughly 50–74% of tables clearly support the same business capability)
- ☐ Good (Roughly 75–89% of tables clearly support the same business capability)
- ☐ Excellent (90–100% of tables clearly support the same business capability)

7. **Domain granularity:** A domain has appropriate granularity when its size matches * the scope of a single business responsibility. This can happen in two ways: a single-table domain is appropriate when that table is genuinely self-contained and independent of other domains, while a multi-table domain is appropriate when all its tables jointly support one coherent business responsibility that could realistically be owned by a single team. Conversely, a domain has inappropriate granularity when it is over-fragmented (splitting off a table that actually belongs with related tables elsewhere) or over-merged (grouping tables with clearly different responsibilities into the same domain). Considering all the inferred domains across the selected data lakes, on average, approximately what proportion have an appropriate granularity?

Contrassegna solo un ovale.

- ☐ Very poor (Fewer than 25% of domains have appropriately matched size; most appear artificially fragmented or merged)
- ☐ Poor (Roughly 25–49% of domains have appropriately matched size)
- ☐ Fair (Roughly 50–74% of domains have appropriately matched size)
- ☐ Good (Roughly 75–89% of domains have appropriately matched size)
- ☐ Excellent (90–100% of domains have a size that appropriately matches the scope of their responsibility)

8. **Practical usability:** A domain is practically usable if an organization could adopt it * as a starting point for assigning real ownership to a business team, without requiring substantial rework (e.g., manually re-splitting, merging, or renaming domains before they could be used). Considering all the inferred domains across all selected data lakes, on average, approximately what proportion could be adopted as-is, with little to no rework, as a starting point for real-world data ownership?

Contrassegna solo un ovale.

- ☐ Very poor (Fewer than 25% of domains could be adopted as-is)
- ☐ Poor (Roughly 25–49% of domains could be adopted as-is)
- ☐ Fair (Roughly 50–74% of domains could be adopted as-is)
- ☐ Good (Roughly 75–89% of domains could be adopted as-is)
- ☐ Excellent (90–100% of domains could be adopted as-is, with little to no rework needed)

9. **Label accuracy:** A business domain label is considered accurate if it correctly * reflects the shared business purpose of the tables in the domain. After reading the label, you should be able to infer the domain's main responsibility without it omitting a major activity or describing activities that the tables do not support. Considering all the inferred domains across all selected data lakes, on average, approximately what proportion have an accurate and representative label?

Contrassegna solo un ovale.

- ☐ Very poor (The label correctly describes fewer than 25% of the tables in this domain)
- ☐ Poor (The label correctly describes roughly 25–49% of the tables in this domain)
- ☐ Fair (The label correctly describes roughly 50–74% of the tables in this domain)
- ☐ Good (The label correctly describes roughly 75–89% of the tables in this domain)
- ☐ Excellent (The label correctly describes 90–100% of the tables in this domain)

10. **Overall model ability:** Overall, considering all selected data lakes, how would you rate the model's ability to infer business domains and assign them meaningful labels? *

Contrassegna solo un ovale.

☐ Very poor

☐ Poor

☐ Fair

☐ Good

☐ Excellent

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